Team 6
Agamemnon

Sched HW Part 3 Report

Overview

For the dynamic scheduler, I created a dynamic scheduler following the scheduler described in idea 2 in part 3 of the assignment. My dynamic scheduler changed the frequencies in which the light and raise moisture behaviors were enabled based on how often the target light level and moisture levels were not in range.

I chose the light behavior because it is a very important behavior as getting a good light range is very important for plant growth. I also noticed that during growth period A the light level was very dependent on the states of the LEDs. Due to the terrabot being in the REL, there was very little outside light coming in and the LEDs greatly outshone the outside light. So whenever the LEDs were on, the light level would reach its target fairly accurately but it would fall far out of target whenever they were off. Thus, by changing the enable schedule of the LEDs, I could significantly change the overall amount of light the plants were getting.

I also chose the raise moisture behavior because it is very important, as I think underwatering was a big issue in growth period A. Since the next growth period has limited water, it is probably a good idea to carefully space the pump actuations so we don’t run out of water.

Dynamic Scheduler Operation

In the dynamic scheduler, I created a monitor to run every 30 minutes and check if currently the light and raise moisture behaviors were not supposed to be scheduled, and if at the same time the light and moisture levels were not in the limit range. I then divided the number of times this was the case by the total number of times the monitor checked throughout the day to get a percentage of missed light and moisture levels for the day.

At the end of the day, I increased the total amount of time the light and raised moisture behaviors were active if the percentage was high, and decreased the total amount of time if the percentage was low. I did this by multiplying the current scheduled hours for the light and raise moisture behaviors by 1/(1.3 - 2*miss percentage), in order to hopefully set the scheduled hours to be somewhat proportional to a number of hours

that could cover the total time required to not have any misses. I then clamped the output within a range for both behaviors.

After getting a new target scheduled hours for both behaviors, I just changed the scheduled hours for both in behavior_info and tried to solve for a schedule that met the requirements. If the solver failed to find a schedule that worked initially, I decreased the light behavior scheduled hours by 4 and decreased the raise moisture scheduled hours by 2 and tried again. If it failed a second time, I gave up and kept the old schedule. I think this would work since the new light and raise moisture scheduled hours are unlikely to change by more than 4 and 2 hours anyways.

Generated Schedules

The schedules generated by the dynamic scheduler were similar to the original schedule. The generated schedules had more light behavior enable intervals and more raise moisture behavior intervals due to the requirements given to the schedule solver. To compensate for the increased number of intervals for these 2 behaviors and the increased time used by these 2 behaviors, there are less lower humid behavior intervals and total time. This may have something to do with the fact that lower humid is the behavior with the second most enable intervals originally, although after the decrease lower humid still had the second most intervals.

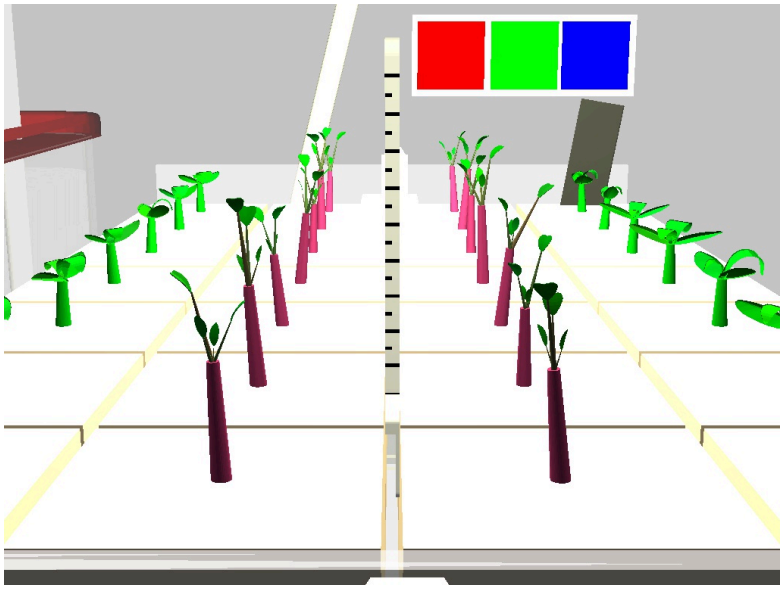
Test Results

I ran both the dynamic scheduler and the normal scheduler for 2 weeks on the Terrabot Docker.

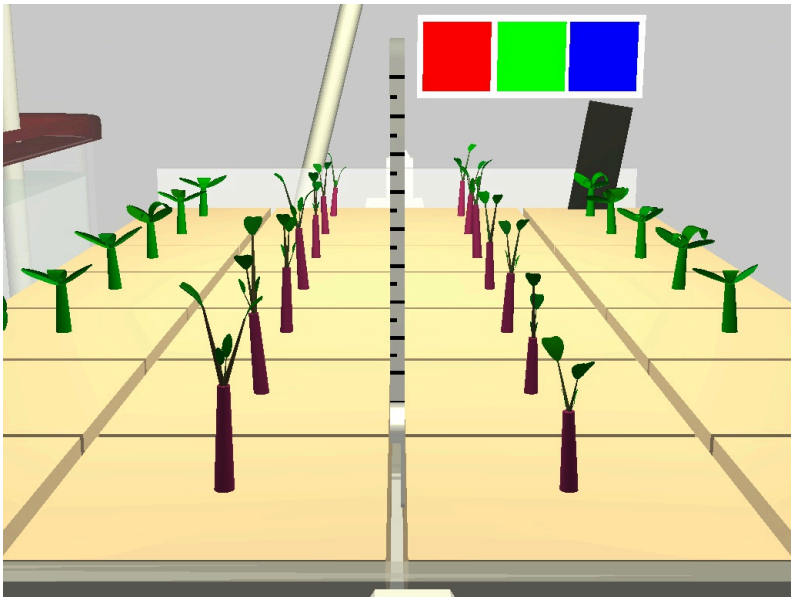
During the 2 week run with the dynamic scheduler, I found that for the light behavior, the scheduled hours quickly increased from their default of 10 to the maximum allowed hours of 14 and stayed there. This was because whenever the light behavior was not running, the light level was likely not in the limit range and so the miss percentage for the lights was always high.

In contrast, the raise moisture behavior scheduled hours went from the starting value of 2 to the minimum of 1 and stayed there. This was likely because after watering, the moisture level stayed within the limit range until the next raise moisture behavior session and the miss rate for the raise moisture behavior was always 0. This shows that in the simulation, the rate of moisture loss is slow enough that running the raise moisture behavior for 1 hour a day is enough to keep the moisture level within range.

At the end of the run, the radish grown by the dynamic scheduler appeared to be a bit larger than the plants grown with the normal schedule. Meanwhile, the lettuce growth by both programs was almost the same. This could be explained by the dynamic scheduler giving the plants more light overall, leading to higher growth in the radish which are more affected by light levels, and not much change in the lettuce which are less affected by light levels.



Dynamic scheduler plants at 1 week



Normal scheduler plants at 1 week